Searching for Boundaries: A Novel Visual Search Task

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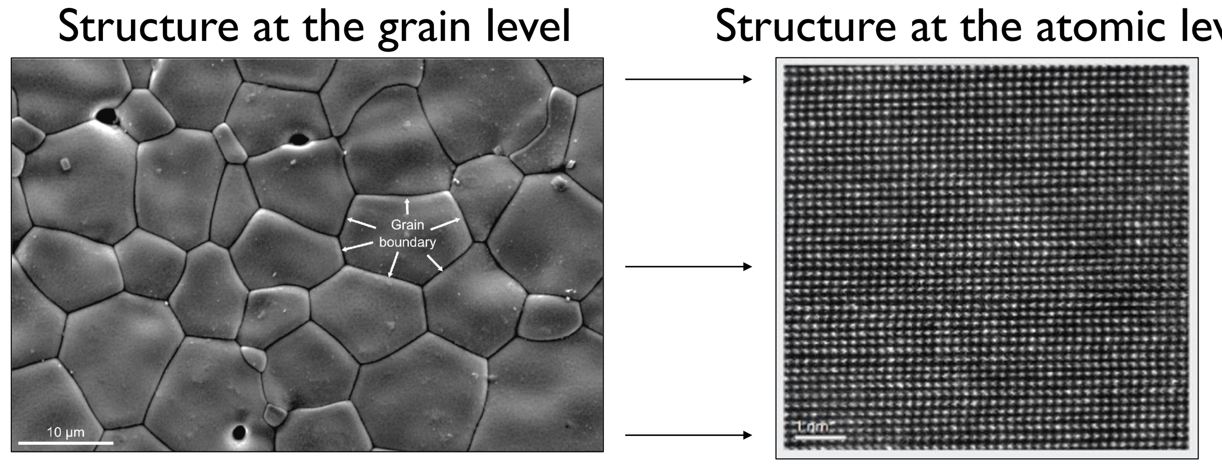
Nano | Human INTERFACES

| Motivation |

- Microscopists examining an atomic-level structure must first view the sample at a lower magnification to find an appropriate grain boundary for higher resolution imaging.

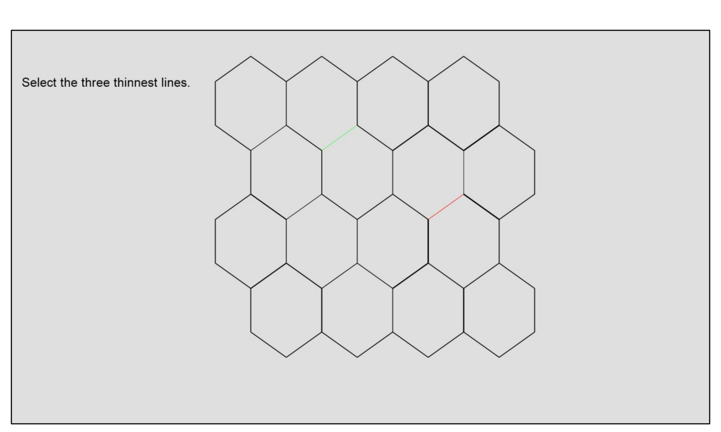
Structure at the grain level — Structure at the atomic level



- Experts use multiple features to identify target boundaries, with the width being a key factor in the decision process.
- The goal of this search process is to identify the best subset of viable boundaries for further imaging.

| Current Project |

- We developed a laboratory analog of the microscopists' work: the ***grain boundary search task.***
- Participants search arrays of hexagons to identify the three thinnest boundaries.



Participants use a mouse to select targets on an interactive display.

Lines turn red when hovered over and green when selected.

- Unique search characteristics of this microscopy task:
  - Multiple potential targets may meet search criteria.
  - Targets vary continuously with non-targets on the critical dimension of width.
  - Targets may differ in width from each other, while also being the same width as unselected targets.
  - Targets are the gaps between grains within a larger structure, rather than the grains themselves.

| Methods |

Task
Select three of the thinnest lines in display

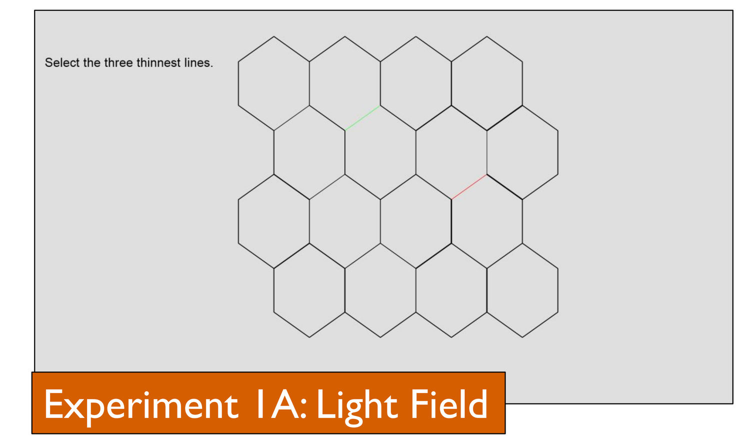
Stimuli
4 x 4 hexagon array with 33 target locations consisting of all interior boundaries

Boundary lines varied randomly from 16 sample widths

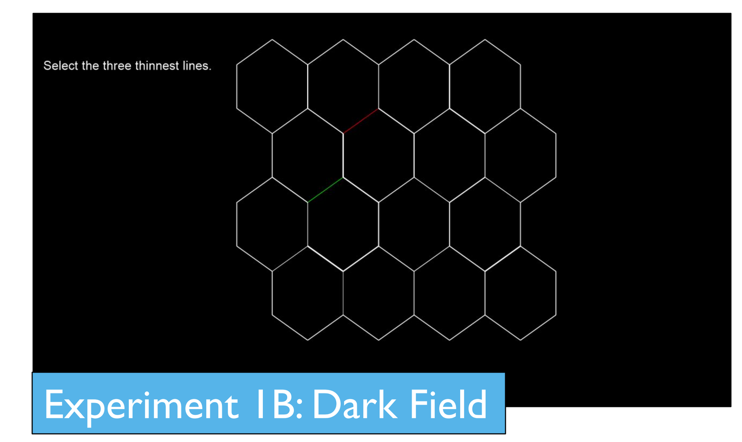
Procedure
Trial sequence:
preview → initiate selection phase → dynamic selection and deselection process → submit response

100 search trials

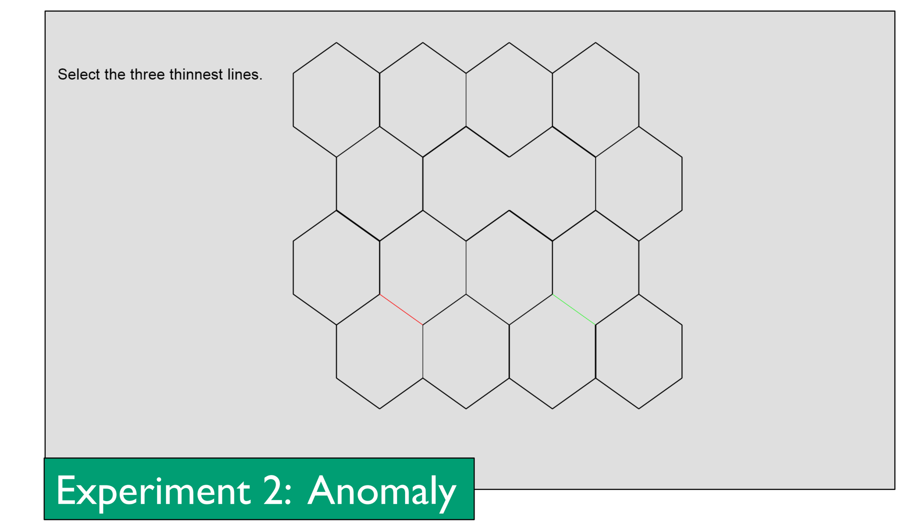
Experiments
Exp. 1A: N = 31
Exp. 1B: N = 29
Exp. 2: N = 32



Experiment 1A: Light Field



Experiment 1B: Dark Field

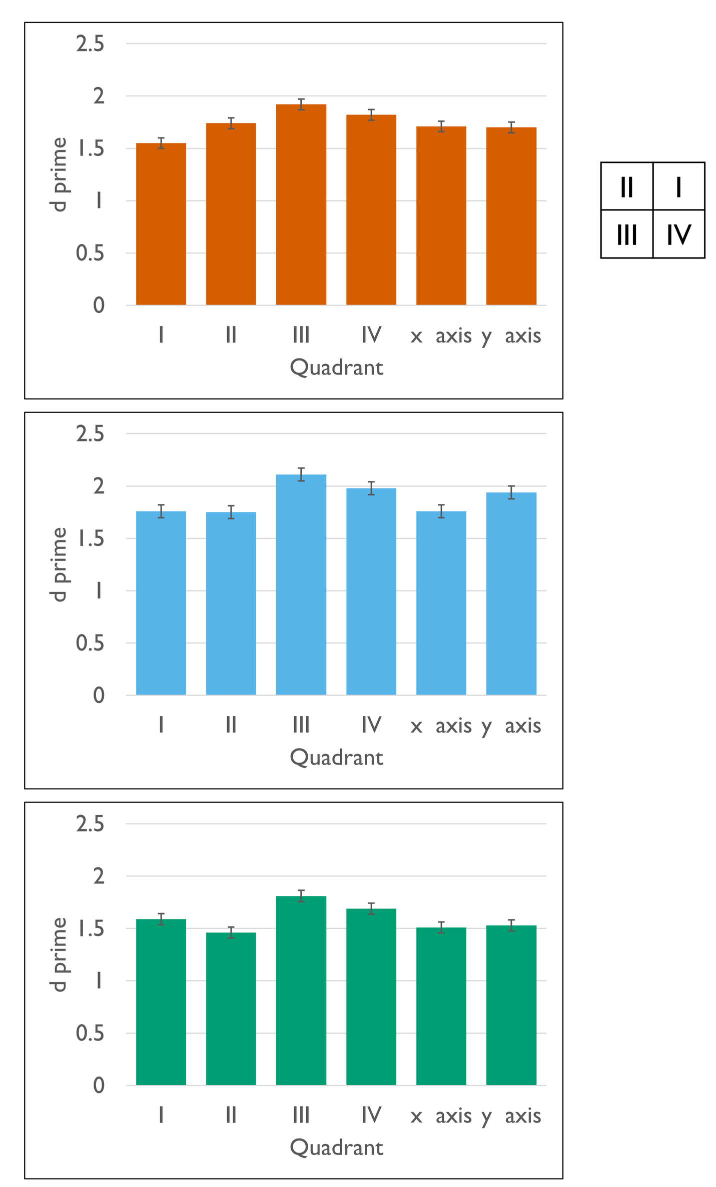


Experiment 2: Anomaly

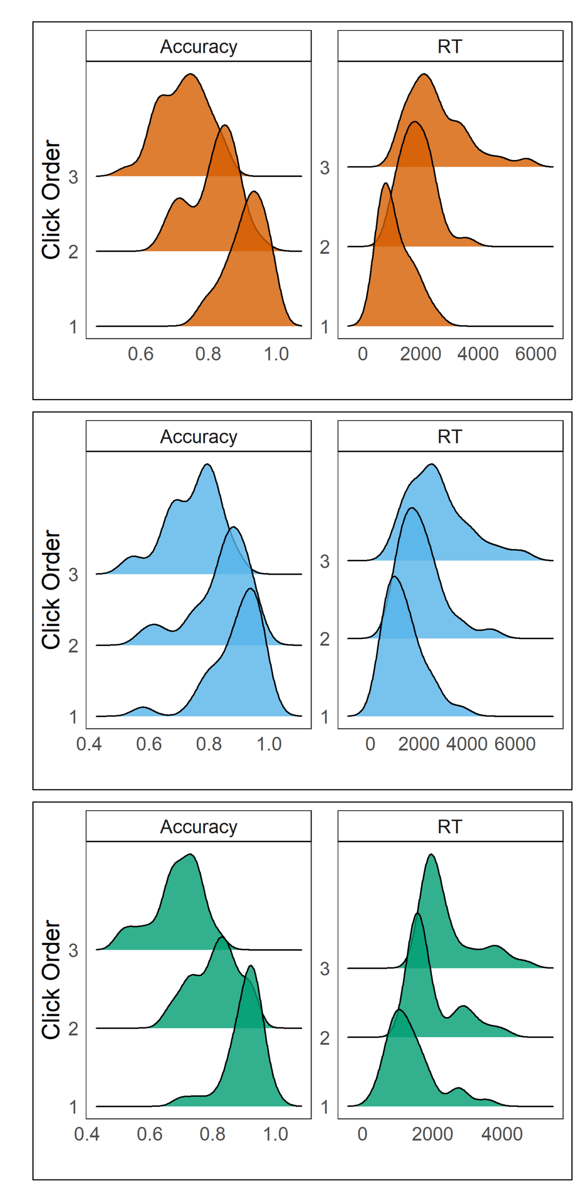
- Natural samples often include anomalies such as unusual grain sizes or imperfections in the sample.
- Experiment 2 introduced an anomaly into 30% of the trials generated by removing a single boundary line.
- If anomalies attract attention, then line boundaries on the anomalous grain may be more likely to be selected.

| Results |

Does selection vary as a function of position in the array?



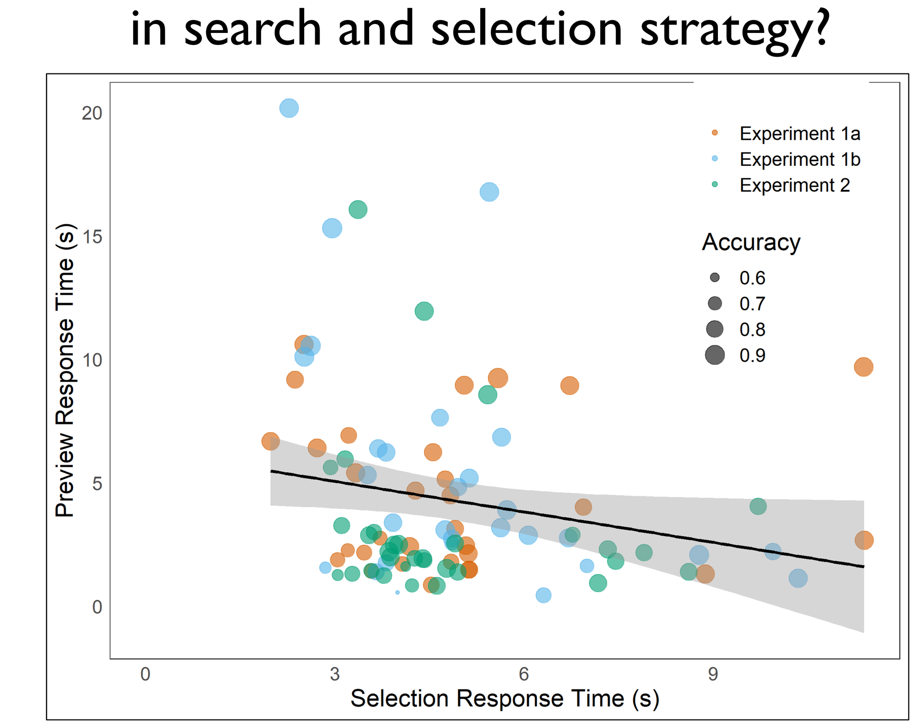
Does selection behavior change across the targets in a trial?



Did the anomalous grain influence search behavior?

	Proportion Selected	Preview RT (sec)	Selection RT (sec)
Anomaly	0.138	3.27	3.65
Non-anomaly	0.145 n.s.	3.10 n.s.	3.45 *

Are there individual differences in search and selection strategy?



Conclusion

- The novel grain boundary search task captures complex search behavior of multiple targets over location and time.
- Current results show both stimulus effects and individual differences in search performance.
- Future studies using the grain boundary search task can explore questions of stimulus characteristics and user expertise that exist in real-world microscopy settings.